

C++ MCQs

1. What will be the output of the following program?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;
    cout << x++ << " " << ++x;
    return 0;
}
```

- A) 10 11
- B) 11 12
- C) 10 12
- D) 11 11

Answer: C

2. Which feature of C++ supports runtime polymorphism?

- A) Function Overloading
- B) Operator Overloading
- C) Virtual Functions
- D) Templates

Answer: C

3. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 5, b = 10;
    cout << (a > b ? a : b);
    return 0;
}
```

- A) 5
- B) 10
- C) True
- D) False

Answer: B

4. Which of the following is not a valid access specifier in C++?

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- A) private
- B) protected
- C) public
- D) internal

Answer: D

5. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
    static int x;
public:
    Test() { x++; }
    static void show() { cout << x; }
};

int Test::x = 0;

int main() {
    Test t1, t2, t3;
    Test::show();
}
```

- A) 0
- B) 1
- C) 2
- D) 3

Answer: D

6. Which operator cannot be overloaded in C++?

- A) []
- B) ()
- C) ::
- D) ++

Answer: C

7. What will be the output?

```
#include <iostream>
using namespace std;
```

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```
int main() {  
    int a = 4;  
    cout << (a << 1);  
}
```

- A) 4
- B) 8
- C) 16
- D) 2

Answer: B

8. What is the size of an empty class in C++?

- A) 0
- B) 1
- C) 2
- D) Depends on compiler

Answer: B

9. Which concept allows one class to acquire properties of another class?

- A) Encapsulation
- B) Abstraction
- C) Inheritance
- D) Overloading

Answer: C

10. What will be the output?

```
#include <iostream>  
using namespace std;  
  
void fun(int &x) {  
    x = x + 5;  
}  
  
int main() {  
    int a = 10;  
    fun(a);  
    cout << a;  
}
```

- A) 10
- B) 15

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- C) 5
- D) Garbage Value

Answer: B

11. Which loop is guaranteed to execute at least once?

- A) for
- B) while
- C) do-while
- D) nested loop

Answer: C

12. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int arr[3] = {1,2,3};
    cout << *(arr + 1);
}
```

- A) 1
- B) 2
- C) 3
- D) Address

Answer: B

13. Which keyword is used to prevent inheritance?

- A) static
- B) final
- C) const
- D) sealed

Answer: B

14. What will happen if memory allocated using `new` is not deleted?

- A) Syntax Error

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- B) Runtime Error
- C) Memory Leak
- D) Compilation Error

Answer: C

15. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() { cout << "A "; }
    ~A() { cout << "B "; }
};

int main() {
    A obj;
}
```

- A) A
- B) B
- C) A B
- D) B A

Answer: C

16. Which of the following supports generic programming?

- A) Inheritance
- B) Templates
- C) Pointers
- D) Constructors

Answer: B

17. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;
    int *p = &x;
    cout << *p;
}
```

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- A) Address of x
- B) 5
- C) Garbage Value
- D) Compiler Error

Answer: B

18. Which type of inheritance may cause ambiguity?

- A) Single
- B) Multiple
- C) Hierarchical
- D) Multilevel

Answer: B

19. What will be the output?

```
#include <iostream>
using namespace std;

class Base {
public:
    virtual void show() {
        cout << "Base";
    }
};

class Derived : public Base {
public:
    void show() {
        cout << "Derived";
    }
};

int main() {
    Base *b;
    Derived d;
    b = &d;
    b->show();
}
```

- A) Base
- B) Derived
- C) Compiler Error
- D) Runtime Error

Answer: B

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20. Which keyword is used to handle exceptions?

- A) error
- B) catch
- C) throw
- D) try

Answer: B

21. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 10;
    cout << sizeof(a++);
}
```

- A) 4
- B) 10
- C) 11
- D) Compilation Error

Answer: A

22. Which STL container stores unique keys in sorted order?

- A) vector
- B) queue
- C) map
- D) stack

Answer: C

23. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    Test() {
        cout << "Constructor ";
    }
    ~Test() {
        cout << "Destructor";
    }
}
```

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```
    }  
};  
  
int main() {  
    Test t;  
}
```

- A) Constructor
- B) Destructor
- C) Constructor Destructor
- D) Destructor Constructor

Answer: C

24. Which cast is safest for downcasting?

- A) static_cast
- B) reinterpret_cast
- C) dynamic_cast
- D) const_cast

Answer: C

25. What will be the output?

```
#include <iostream>  
using namespace std;  
  
inline int square(int x) {  
    return x * x;  
}  
  
int main() {  
    cout << square(5);  
}
```

- A) 5
- B) 10
- C) 20
- D) 25

Answer: D

26. Which operator is used to access class members through pointer?

- A) .
- B) ::

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- C) ->
- D) *

Answer: C

27. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 0;
    while(x < 3) {
        cout << x;
        x++;
    }
}
```

- A) 012
- B) 123
- C) 321
- D) Infinite Loop

Answer: A

28. Which function is automatically called when an object is copied?

- A) Destructor
- B) Friend Function
- C) Copy Constructor
- D) Virtual Function

Answer: C

29. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    virtual void show() = 0;
};

int main() {
    A obj;
}
```

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- A) No Output
- B) Runtime Error
- C) Compilation Error
- D) Garbage Value

Answer: C

30. What will be the output?

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> v = {1,2,3};
    v.push_back(4);
    cout << v.size();
}
```

- A) 3
- B) 4
- C) 5
- D) Error

Answer: B

31. Which keyword is used to define a constant member function?

- A) constant
- B) immutable
- C) final
- D) const

Answer: D

32. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 5;
    int b = ++a + a++;
    cout << b;
}
```

- A) 11

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- B) 12
- C) 13
- D) Undefined Behavior

Answer: D

33. Which header file is required for file handling?

- A) string
- B) iostream
- C) fstream
- D) stdlib

Answer: C

34. What will be the output?

```
#include <iostream>
using namespace std;

template <class T>
T add(T a, T b) {
    return a + b;
}

int main() {
    cout << add(2,3);
}
```

- A) 23
- B) 5
- C) Error
- D) Garbage Value

Answer: B

35. Which function is invoked automatically before object destruction?

- A) Constructor
- B) Friend Function
- C) Destructor
- D) Main Function

Answer: C

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36. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int arr[2][2] = {{1,2},{3,4}};
    cout << arr[1][0];
}
```

- A) 1
- B) 2
- C) 3
- D) 4

Answer: C

37. Which of the following is compile-time polymorphism?

- A) Virtual Function
- B) Function Overriding
- C) Function Overloading
- D) Dynamic Binding

Answer: C

38. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "Hello";
    }
};

int main() {
    A a1, a2;
}
```

- A) Hello
- B) HelloHello
- C) Error
- D) No Output

Answer: B

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39. Which storage class retains value between function calls?

- A) auto
- B) register
- C) static
- D) extern

Answer: C

40. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    try {
        throw 10;
    }
    catch(int x) {
        cout << x;
    }
}
```

- A) Error
- B) 0
- C) 10
- D) Garbage Value

Answer: C

41. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    Test() {
        cout << "A";
    }
    Test(const Test&) {
        cout << "B";
    }
};

int main() {
    Test t1;
    Test t2 = t1;
}
```

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- A) A
- B) AB
- C) ABB
- D) B

Answer: B

42. Which of the following correctly declares a pointer to a function?

- A) `int *func();`
- B) `int (*func)();`
- C) `int &func();`
- D) `pointer int func();`

Answer: B

43. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 5;
    cout << a+++a;
}
```

- A) 10
- B) 11
- C) 12
- D) Compilation Error

Answer: B

44. Which of the following is true about references in C++?

- A) References can be null
- B) References can be reassigned
- C) References must be initialized when declared
- D) References occupy no memory

Answer: C

45. What will be the output?

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```
#include <iostream>
using namespace std;

class A {
public:
    virtual void show() {
        cout << "A";
    }
};

class B : public A {
public:
    void show() override {
        cout << "B";
    }
};

int main() {
    A *ptr = new B();
    ptr->show();
    delete ptr;
}
```

- A) A
- B) B
- C) AB
- D) Compilation Error

Answer: B

46. Which operator is overloaded automatically for every class by default?

- A) +
- B) []
- C) =
- D) <<

Answer: C

47. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;
    int &y = x;
    y++;
    cout << x;
}
```

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- A) 5
- B) 6
- C) Garbage Value
- D) Error

Answer: B

48. Which keyword is used to inherit a class publicly?

- A) extends
- B) inherits
- C) public
- D) protected

Answer: C

49. What will be the output?

```
#include <iostream>
using namespace std;

class Base {
public:
    Base() {
        cout << "Base ";
    }
};

class Derived : public Base {
public:
    Derived() {
        cout << "Derived";
    }
};

int main() {
    Derived d;
}
```

- A) Derived Base
- B) Base Derived
- C) Base
- D) Derived

Answer: B

50. Which STL container provides fast insertion/removal at both ends?

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- A) vector
- B) stack
- C) deque
- D) map

Answer: C

51. What will be the output?

```
#include <iostream>
using namespace std;

class A {
    int x;
public:
    A(int a) : x(a) {}
    void show() const {
        cout << x;
    }
};

int main() {
    A obj(10);
    obj.show();
}
```

- A) 0
- B) 10
- C) Garbage Value
- D) Error

Answer: B

52. Which of the following cannot have virtual functions?

- A) Base Class
- B) Derived Class
- C) Static Member Function
- D) Abstract Class

Answer: C

53. What will be the output?

```
#include <iostream>
using namespace std;
```

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```
int main() {  
    int *p = new int(25);  
    cout << *p;  
    delete p;  
}
```

- A) Address
- B) 25
- C) Garbage Value
- D) Error

Answer: B

54. Which of the following is used for exception handling hierarchy?

- A) namespace
- B) class inheritance
- C) templates
- D) operator overloading

Answer: B

55. What will be the output?

```
#include <iostream>  
using namespace std;  
  
class A {  
public:  
    void show() {  
        cout << "A";  
    }  
};  
  
class B : public A {  
public:  
    void show() {  
        cout << "B";  
    }  
};  
  
int main() {  
    B obj;  
    obj.A::show();  
}
```

- A) A
- B) B
- C) AB
- D) Error

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Answer: A

56. Which keyword prevents a function from being overridden?

- A) virtual
- B) static
- C) final
- D) const

Answer: C

57. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 3;
    switch(x) {
        case 1:
        case 2:
        case 3:
            cout << "Three";
            break;
        default:
            cout << "Default";
    }
}
```

- A) One
- B) Two
- C) Three
- D) Default

Answer: C

58. Which of the following is not part of STL?

- A) Containers
- B) Iterators
- C) Algorithms
- D) Compiler

Answer: D

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59. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    ~A() {
        cout << "Destructor";
    }
};

int main() {
    A *ptr = new A();
    delete ptr;
}
```

- A) No Output
- B) Destructor
- C) Error
- D) Garbage Value

Answer: B

60. Which of the following creates ambiguity in multiple inheritance?

- A) Same constructor names
- B) Same variable names in base classes
- C) Different access specifiers
- D) Different namespaces

Answer: B

61. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int arr[] = {10,20,30};
    cout << *(arr + 2);
}
```

- A) 10
- B) 20
- C) 30
- D) Address

Answer: C

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62. Which keyword is used to define an abstract class?

- A) abstract
- B) interface
- C) virtual
- D) No special keyword

Answer: D

63. What will be the output?

```
#include <iostream>
using namespace std;

void func(int x = 5) {
    cout << x;
}

int main() {
    func();
}
```

- A) 0
- B) 5
- C) Garbage Value
- D) Error

Answer: B

64. Which file mode truncates existing file content?

- A) ios::app
- B) ios::binary
- C) ios::trunc
- D) ios::ate

Answer: C

65. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
```

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```
int x;  
Test() {  
    x = 100;  
}  
};  
  
int main() {  
    Test t;  
    cout << t.x;  
}
```

- A) Garbage Value
- B) 0
- C) 100
- D) Error

Answer: C

66. Which type of constructor takes another object of same class as argument?

- A) Default Constructor
- B) Parameterized Constructor
- C) Copy Constructor
- D) Dynamic Constructor

Answer: C

67. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int x = 1;  
    for(int i = 0; i < 3; i++) {  
        x *= 2;  
    }  
    cout << x;  
}
```

- A) 2
- B) 4
- C) 8
- D) 16

Answer: C

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68. Which keyword is used for namespaces?

- A) package
- B) namespace
- C) module
- D) define

Answer: B

69. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    virtual void display() {
        cout << "A";
    }
};

class B : public A {
public:
    void display() {
        cout << "B";
    }
};

int main() {
    A a;
    B b;
    A *ptr[2] = {&a, &b};

    ptr[0]->display();
    ptr[1]->display();
}
```

- A) AA
- B) AB
- C) BB
- D) BA

Answer: B

70. Which of the following is true about templates?

- A) Templates support code reusability
- B) Templates increase code duplication
- C) Templates work only with integers
- D) Templates cannot be used with classes

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Answer: A

71. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;
    int *ptr = &x;
    *ptr += 5;
    cout << x;
}
```

- A) 5
- B) 10
- C) 15
- D) Error

Answer: C

72. Which of the following is a sequence container?

- A) map
- B) set
- C) vector
- D) unordered_map

Answer: C

73. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "1";
    }
    ~A() {
        cout << "2";
    }
};

int main() {
    A obj1;
    {
        A obj2;
```


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```
}  
}
```

- A) 1122
- B) 1212
- C) 1221
- D) 2112

Answer: A

74. Which operator is used to access members of namespace?

- A) .
- B) ->
- C) ::
- D) *

Answer: C

75. What will be the output?

```
#include <iostream>  
using namespace std;  
  
class Base {  
public:  
    virtual ~Base() {  
        cout << "Base";  
    }  
};  
  
class Derived : public Base {  
public:  
    ~Derived() {  
        cout << "Derived";  
    }  
};  
  
int main() {  
    Base *ptr = new Derived();  
    delete ptr;  
}
```

- A) Base
- B) Derived
- C) DerivedBase
- D) BaseDerived

Answer: C

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76. Which STL container is implemented using balanced trees?

- A) vector
- B) queue
- C) map
- D) stack

Answer: C

77. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    static int count;
    Test() {
        count++;
    }
};

int Test::count = 0;

int main() {
    Test t1, t2;
    cout << Test::count;
}
```

- A) 0
- B) 1
- C) 2
- D) Error

Answer: C

78. Which cast performs low-level reinterpretation of bits?

- A) dynamic_cast
- B) static_cast
- C) reinterpret_cast
- D) const_cast

Answer: C

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79. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 5;
    cout << (a == 5 && a < 10);
}
```

- A) 0
- B) 1
- C) True
- D) False

Answer: B

80. Which of the following is true about destructors?

- A) They can be overloaded
- B) They can take arguments
- C) Only one destructor is allowed per class
- D) They return values

Answer: C

81. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    Test() {
        cout << "C";
    }
    ~Test() {
        cout << "D";
    }
};

int main() {
    Test t1;
    Test t2;
}
```

- A) CCDD
- B) CDCD
- C) CCDD
- D) CDDC

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Answer: A

82. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;
    cout << x++ + ++x;
}
```

- A) 10
- B) 11
- C) 12
- D) Undefined Behavior

Answer: D

83. Which of the following is true for virtual functions?

- A) They can be static
- B) They are resolved at compile time
- C) They support dynamic binding
- D) They cannot be inherited

Answer: C

84. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    void show() {
        cout << "A";
    }
};

class B : public A {
};

int main() {
    B obj;
    obj.show();
}
```

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- A) A
- B) B
- C) AB
- D) Error

Answer: A

85. Which keyword is used to allocate memory dynamically for arrays?

- A) malloc
- B) alloc
- C) new
- D) create

Answer: C

86. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int arr[5] = {1,2};
    cout << arr[3];
}
```

- A) Garbage Value
- B) 0
- C) 2
- D) Error

Answer: B

87. Which of the following is not a pillar of OOP?

- A) Encapsulation
- B) Abstraction
- C) Compilation
- D) Polymorphism

Answer: C

88. What will be the output?

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```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "A";
    }
};

class B : public A {
public:
    B() {
        cout << "B";
    }
};

class C : public B {
public:
    C() {
        cout << "C";
    }
};

int main() {
    C obj;
}
```

- A) ABC
- B) CBA
- C) BAC
- D) ACB

Answer: A

89. Which operator is used to determine size of datatype or variable?

- A) length
- B) size
- C) sizeof
- D) datatype

Answer: C

90. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;
    int *p = &x;
```

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```
int **q = &p;

cout << **q;
}
```

- A) Address of x
- B) Address of p
- C) 10
- D) Garbage Value

Answer: C

91. Which STL container allows duplicate elements?

- A) set
- B) map
- C) multiset
- D) unordered_map

Answer: C

92. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
    int x;
public:
    Test(int a) {
        x = a;
    }

    Test(Test &t) {
        x = t.x;
        cout << x;
    }
};

int main() {
    Test t1(50);
    Test t2(t1);
}
```

- A) 0
- B) 50
- C) Garbage Value
- D) Error

Answer: B

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93. Which keyword is used to inherit privately?

- A) private
- B) protected
- C) friend
- D) virtual

Answer: A

94. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 1;

    if(x)
        cout << "True";
    else
        cout << "False";
}
```

- A) True
- B) False
- C) Error
- D) No Output

Answer: A

95. Which function is called automatically when object goes out of scope?

- A) Constructor
- B) Friend Function
- C) Destructor
- D) Inline Function

Answer: C

96. What will be the output?

```
#include <iostream>
using namespace std;

class A {
```


C++ MCQs

```
public:
    virtual void show() {
        cout << "A";
    }
};

class B : public A {
public:
    void show() {
        cout << "B";
    }
};

int main() {
    A obj;
    B obj2;

    A *ptr = &obj2;
    ptr->show();
}
```

- A) A
- B) B
- C) AB
- D) Error

Answer: B

97. Which of the following containers is adapter container?

- A) vector
- B) list
- C) stack
- D) array

Answer: C

98. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    static void display() {
        cout << "Static";
    }
};

int main() {
    Test::display();
}
```

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- A) Static
- B) Error
- C) No Output
- D) Garbage Value

Answer: A

99. Which keyword is used to stop exception propagation?

- A) break
- B) throw
- C) catch
- D) terminate

Answer: C

100. What will be the output?

```
#include <iostream>
using namespace std;

template <typename T>
T maximum(T a, T b) {
    return (a > b) ? a : b;
}

int main() {
    cout << maximum(10, 20);
}
```

- A) 10
- B) 20
- C) 0
- D) Error

Answer: B

101. Which of the following is true about constructors?

- A) Constructors can be virtual
- B) Constructors can return values
- C) Constructors are automatically invoked
- D) Constructors cannot be overloaded

Answer: C

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102. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 4;

    switch(x) {
        case 1:
            cout << "One";
            break;
        case 4:
            cout << "Four";
            break;
        default:
            cout << "Default";
    }
}
```

- A) One
- B) Four
- C) Default
- D) Error

Answer: B

103. Which concept is achieved using function overriding?

- A) Compile-time polymorphism
- B) Runtime polymorphism
- C) Encapsulation
- D) Abstraction

Answer: B

104. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    char ch = 'A';
    cout << int(ch);
}
```

- A) A
- B) 64

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- C) 65
- D) Error

Answer: C

105. Which header file is required for `std::vector`?

- A) stack
- B) vector
- C) map
- D) queue

Answer: B

106. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    int x = 10;
};

int main() {
    A obj;
    A *ptr = &obj;

    cout << ptr->x;
}
```

- A) 0
- B) 10
- C) Garbage Value
- D) Error

Answer: B

107. Which inheritance type combines more than one inheritance?

- A) Hierarchical
- B) Hybrid
- C) Multilevel
- D) Single

Answer: B

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108. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    Test() {
        cout << "1";
    }

    ~Test() {
        cout << "2";
    }
};

int main() {
    Test t1;
    {
        Test t2;
    }
    cout << "3";
}
```

- A) 11223
- B) 11322
- C) 12123
- D) 12312

Answer: B

109. Which keyword is used to define constant objects?

- A) static
- B) immutable
- C) final
- D) const

Answer: D

110. What will be the output?

```
#include <iostream>
using namespace std;

class Base {
public:
    virtual void display() {
        cout << "Base";
    }
}
```

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```
    }  
};  
  
class Derived : public Base {  
public:  
    void display() {  
        cout << "Derived";  
    }  
};  
  
int main() {  
    Derived d;  
    Base &b = d;  
    b.display();  
}
```

- A) Base
- B) Derived
- C) Error
- D) No Output

Answer: B

111. Which function is used to open a file in C++?

- A) fopen()
- B) open()
- C) fileopen()
- D) start()

Answer: B

112. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int x = 5;  
  
    while(x--) {  
        cout << x;  
    }  
}
```

- A) 54321
- B) 43210
- C) 01234
- D) Infinite Loop

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Answer: B

113. Which STL container stores elements in contiguous memory?

- A) list
- B) map
- C) vector
- D) set

Answer: C

114. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "A";
    }
};

int main() {
    A arr[2];
}
```

- A) A
- B) AA
- C) Error
- D) No Output

Answer: B

115. Which operator is overloaded for extraction operation?

- A) <<
- B) >>
- C) ==
- D) +

Answer: B

116. What will be the output?

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```
#include <iostream>
using namespace std;

class Test {
public:
    void display() const {
        cout << "Const Function";
    }
};

int main() {
    const Test t;
    t.display();
}
```

- A) Error
- B) Const Function
- C) No Output
- D) Garbage Value

Answer: B

117. Which of the following supports data hiding?

- A) Abstraction
- B) Encapsulation
- C) Inheritance
- D) Polymorphism

Answer: B

118. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 10;
    int b = 20;

    cout << ((a > b) ? a : b);
}
```

- A) 10
- B) 20
- C) True
- D) False

Answer: B

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119. Which keyword is used to define friend function?

- A) ally
- B) friend
- C) companion
- D) access

Answer: B

120. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 2;

    do {
        cout << x;
        x++;
    } while(x < 5);
}
```

- A) 234
- B) 1234
- C) 2345
- D) Infinite Loop

Answer: A

121. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "A";
    }

    ~A() {
        cout << "B";
    }
};

int main() {
    A obj1;
    {
```

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```
        A obj2;  
    }  
    cout << "C";  
}
```

- A) AABCBB
- B) AABBCB
- C) ABABCC
- D) AABBBC

Answer: B

122. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int x = 5;  
    cout << ++x + x++;  
}
```

- A) 11
- B) 12
- C) 13
- D) Undefined Behavior

Answer: D

123. Which of the following is true about abstract classes?

- A) Objects can be created directly
- B) They must contain only pure virtual functions
- C) They cannot be instantiated
- D) Constructors are not allowed

Answer: C

124. What will be the output?

```
#include <iostream>  
using namespace std;  
  
class Base {  
public:  
    void show() {  
        cout << "Base";  
    }  
}
```

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```
    }  
};  
  
class Derived : public Base {  
};  
  
int main() {  
    Derived d;  
    d.show();  
}
```

- A) Base
- B) Derived
- C) Error
- D) No Output

Answer: A

125. Which operator is used to access a class member using object?

- A) ->
- B) ::
- C) .
- D) *

Answer: C

126. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int arr[3] = {5,10,15};  
  
    cout << *(arr);  
}
```

- A) 0
- B) 5
- C) 10
- D) Address

Answer: B

127. Which type of polymorphism is achieved using virtual functions?

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- A) Compile-time
- B) Runtime
- C) Static
- D) Early Binding

Answer: B

128. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "1";
    }
};

class B {
public:
    B() {
        cout << "2";
    }
};

class C : public A, public B {
public:
    C() {
        cout << "3";
    }
};

int main() {
    C obj;
}
```

- A) 123
- B) 321
- C) 213
- D) 132

Answer: A

129. Which of the following correctly declares a constant integer?

- A) `int const x;`
- B) `const int x = 10;`
- C) `constant int x = 10;`
- D) `int final x = 10;`

C++ MCQs

Answer: B

130. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 10;
    int &ref = a;

    ref = 20;

    cout << a;
}
```

- A) 10
- B) 20
- C) Garbage Value
- D) Error

Answer: B

131. Which STL container does not allow random access?

- A) vector
- B) deque
- C) list
- D) array

Answer: C

132. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    Test() {
        cout << "Constructor ";
    }

    ~Test() {
        cout << "Destructor ";
    }
};

int main() {
```

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```
Test t1;  
Test t2;  
}
```

- A) Constructor Constructor Destructor Destructor
- B) Constructor Destructor Constructor Destructor
- C) Destructor Constructor Destructor Constructor
- D) Constructor Constructor

Answer: A

133. Which keyword allows function overriding?

- A) static
- B) virtual
- C) friend
- D) inline

Answer: B

134. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int x = 3;  
  
    if(x > 0)  
        if(x < 5)  
            cout << "A";  
        else  
            cout << "B";  
}
```

- A) A
- B) B
- C) AB
- D) No Output

Answer: A

135. Which function is used to release dynamically allocated array memory?

- A) free[]
- B) remove[]
- C) delete[]

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- D) clear[]

Answer: C

136. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    virtual void display() {
        cout << "A";
    }
};

class B : public A {
public:
    void display() {
        cout << "B";
    }
};

int main() {
    A *ptr = new B;
    ptr->display();
}
```

- A) A
- B) B
- C) Error
- D) No Output

Answer: B

137. Which of the following is used for generic class definitions?

- A) namespace
- B) template
- C) typedef
- D) inline

Answer: B

138. What will be the output?

```
#include <iostream>
using namespace std;
```

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```
int main() {
    int x = 1;

    for(int i = 0; i < 4; i++) {
        x += i;
    }

    cout << x;
}
```

- A) 5
- B) 6
- C) 7
- D) 8

Answer: C

139. Which access specifier allows access within derived classes only?

- A) public
- B) private
- C) protected
- D) internal

Answer: C

140. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    static int x;
};

int Test::x = 100;

int main() {
    cout << Test::x;
}
```

- A) 0
- B) 100
- C) Garbage Value
- D) Error

Answer: B

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141. Which of the following supports compile-time polymorphism?

- A) Virtual Functions
- B) Function Overloading
- C) Function Overriding
- D) Inheritance

Answer: B

142. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;

    while(x > 0) {
        cout << x;
        x--;
    }
}
```

- A) 12345
- B) 54321
- C) 543210
- D) Infinite Loop

Answer: B

143. Which file stream class is used for both reading and writing?

- A) ifstream
- B) ofstream
- C) fstream
- D) ostream

Answer: C

144. What will be the output?

```
#include <iostream>
using namespace std;

class A {
```

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```
public:
    A() {
        cout << "A";
    }

    ~A() {
        cout << "B";
    }
};

int main() {
    A *ptr = new A();
    delete ptr;
}
```

- A) A
- B) B
- C) AB
- D) BA

Answer: C

145. Which keyword is used to throw an exception?

- A) catch
- B) try
- C) throw
- D) exception

Answer: C

146. What will be the output?

```
#include <iostream>
using namespace std;

class Base {
public:
    virtual void show() {
        cout << "Base";
    }
};

class Derived : public Base {
public:
    void show() {
        cout << "Derived";
    }
};

int main() {
    Derived d;
```

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```
Base *ptr = &d;

ptr->show();
}
```

- A) Base
- B) Derived
- C) Error
- D) No Output

Answer: B

147. Which STL container is best suited for frequent insertion/deletion in middle?

- A) vector
- B) array
- C) list
- D) stack

Answer: C

148. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    void display() {
        cout << "A";
    }
};

class B : public A {
};

int main() {
    B obj;
    obj.display();
}
```

- A) A
- B) B
- C) Error
- D) AB

Answer: A

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149. Which operator is used to access pointer member variables?

- A) .
- B) ->
- C) ::
- D) *

Answer: B

150. What will be the output?

```
#include <iostream>
using namespace std;

template <typename T>
T square(T x) {
    return x * x;
}

int main() {
    cout << square(4);
}
```

- A) 4
- B) 8
- C) 16
- D) Error

Answer: C

151. Which keyword is used to avoid name conflicts?

- A) class
- B) namespace
- C) template
- D) typedef

Answer: B

152. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 5;
    int b = 2;
```

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```
    cout << a % b;  
}
```

- A) 2
- B) 1
- C) 0
- D) 5

Answer: B

153. Which concept combines data and methods into single unit?

- A) Inheritance
- B) Abstraction
- C) Encapsulation
- D) Polymorphism

Answer: C

154. What will be the output?

```
#include <iostream>  
using namespace std;  
  
class Test {  
public:  
    Test() {  
        cout << "X";  
    }  
};  
  
int main() {  
    Test t[3];  
}
```

- A) X
- B) XX
- C) XXX
- D) Error

Answer: C

155. Which cast removes constant property from variable?

- A) static_cast
- B) reinterpret_cast

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- C) const_cast
- D) dynamic_cast

Answer: C

156. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 0;

    do {
        cout << x;
        x++;
    } while(x < 3);
}
```

- A) 012
- B) 123
- C) 0123
- D) Infinite Loop

Answer: A

157. Which of the following is an associative container?

- A) vector
- B) deque
- C) map
- D) array

Answer: C

158. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    virtual void fun() = 0;
};

class B : public A {
public:
    void fun() {
```

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```
        cout << "Implemented";
    }
};

int main() {
    B obj;
    obj.fun();
}
```

- A) Error
- B) Implemented
- C) No Output
- D) Garbage Value

Answer: B

159. Which keyword is used to define inline functions?

- A) static
- B) inline
- C) friend
- D) virtual

Answer: B

160. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;

    cout << (x > 5 ? "Yes" : "No");
}
```

- A) Yes
- B) No
- C) Error
- D) Garbage Value

Answer: A

161. What will be the output?

```
#include <iostream>
using namespace std;
```

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```
class A {
public:
    A() {
        cout << "1";
    }
    ~A() {
        cout << "2";
    }
};

int main() {
    A a1;
    {
        A a2;
    }
    cout << "3";
}
```

- A) 11223
- B) 11322
- C) 12123
- D) 12213

Answer: B

162. Which keyword is used to define a class in C++?

- A) object
- B) define
- C) class
- D) struct

Answer: C

163. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;
    int y = x++ + 2;

    cout << y;
}
```

- A) 5
- B) 6
- C) 7
- D) 8

C++ MCQs

Answer: C

164. Which feature allows same function name with different parameter lists?

- A) Function Overriding
- B) Function Overloading
- C) Encapsulation
- D) Inheritance

Answer: B

165. What will be the output?

```
#include <iostream>
using namespace std;

class Base {
public:
    virtual void show() {
        cout << "Base";
    }
};

class Derived : public Base {
public:
    void show() {
        cout << "Derived";
    }
};

int main() {
    Base *ptr = new Derived();
    ptr->show();
}
```

- A) Base
- B) Derived
- C) Error
- D) No Output

Answer: B

166. Which operator is used to allocate dynamic memory in C++?

- A) alloc
- B) malloc
- C) new
- D) create

C++ MCQs

Answer: C

167. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int arr[] = {2,4,6,8};

    cout << *(arr + 3);
}
```

- A) 2
- B) 4
- C) 6
- D) 8

Answer: D

168. Which access specifier makes members accessible only within same class?

- A) public
- B) protected
- C) private
- D) friend

Answer: C

169. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    static int count;

    Test() {
        count++;
    }
};

int Test::count = 0;

int main() {
    Test t1, t2, t3;
```

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```
        cout << Test::count;  
    }
```

- A) 0
- B) 1
- C) 2
- D) 3

Answer: D

170. Which loop is best suited when number of iterations is known?

- A) while
- B) do-while
- C) for
- D) nested while

Answer: C

171. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int x = 1;  
  
    while(x <= 3) {  
        cout << x;  
        x++;  
    }  
}
```

- A) 123
- B) 321
- C) 012
- D) Infinite Loop

Answer: A

172. Which keyword is used to stop loop execution immediately?

- A) continue
- B) return
- C) break
- D) exit

C++ MCQs

Answer: C

173. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "Constructor";
    }
};

int main() {
    A obj;
}
```

- A) Constructor
- B) Destructor
- C) Error
- D) No Output

Answer: A

174. Which concept allows one interface with multiple implementations?

- A) Encapsulation
- B) Polymorphism
- C) Inheritance
- D) Abstraction

Answer: B

175. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;

    cout << (x == 10);
}
```

- A) 10
- B) True
- C) 1

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- D) 0

Answer: C

176. Which function is automatically called when object is created?

- A) Destructor
- B) Constructor
- C) Main Function
- D) Friend Function

Answer: B

177. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    void display() {
        cout << "Display";
    }
};

int main() {
    Test t;
    t.display();
}
```

- A) Display
- B) Error
- C) No Output
- D) Garbage Value

Answer: A

178. Which keyword is used to define pure virtual function?

- A) pure
- B) virtual
- C) abstract
- D) = 0

Answer: D

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179. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 5;

    cout << ++a;
}
```

- A) 5
- B) 6
- C) 7
- D) Error

Answer: B

180. Which type of inheritance uses more than one base class?

- A) Single
- B) Multiple
- C) Multilevel
- D) Hierarchical

Answer: B

181. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    void show() {
        cout << "A";
    }
};

class B : public A {
};

int main() {
    B obj;
    obj.show();
}
```

- A) A

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- B) B
- C) Error
- D) No Output

Answer: A

182. Which operator is used for logical AND?

- A) &
- B) &&
- C) ||
- D) !

Answer: B

183. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 4;

    switch(x) {
        case 1:
            cout << "One";
            break;

        case 4:
            cout << "Four";
            break;
    }
}
```

- A) One
- B) Four
- C) Error
- D) No Output

Answer: B

184. Which keyword is used to define constant variables?

- A) immutable
- B) static
- C) const
- D) final

C++ MCQs

Answer: C

185. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int arr[2][2] = {{1,2},{3,4}};

    cout << arr[0][1];
}
```

- A) 1
- B) 2
- C) 3
- D) 4

Answer: B

186. Which concept hides implementation details from user?

- A) Inheritance
- B) Encapsulation
- C) Abstraction
- D) Polymorphism

Answer: C

187. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;

    if(x > 0)
        cout << "Positive";
    else
        cout << "Negative";
}
```

- A) Positive
- B) Negative
- C) Error
- D) No Output

C++ MCQs

Answer: A

188. Which header file is required for input and output stream?

- A) stdio.h
- B) conio.h
- C) iostream
- D) math.h

Answer: C

189. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    Test() {
        cout << "Hello";
    }

    ~Test() {
        cout << "Bye";
    }
};

int main() {
    Test t;
}
```

- A) Hello
- B) Bye
- C) HelloBye
- D) ByeHello

Answer: C

190. Which STL container follows LIFO principle?

- A) queue
- B) vector
- C) stack
- D) map

Answer: C

C++ MCQs

191. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;
    int *p = &x;

    cout << *p;
}
```

- A) Address
- B) 10
- C) Garbage Value
- D) Error

Answer: B

192. Which keyword is used to create namespaces?

- A) package
- B) namespace
- C) module
- D) include

Answer: B

193. What will be the output?

```
#include <iostream>
using namespace std;

template <typename T>
T add(T a, T b) {
    return a + b;
}

int main() {
    cout << add(5,10);
}
```

- A) 15
- B) 510
- C) Error
- D) 0

C++ MCQs

Answer: A

194. Which keyword is used to catch exceptions?

- A) try
- B) throw
- C) catch
- D) error

Answer: C

195. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;

    do {
        cout << x;
        x--;
    } while(x > 2);
}
```

- A) 543
- B) 5432
- C) 432
- D) Infinite Loop

Answer: A

196. Which operator is used to access members through object pointer?

- A) .
- B) ::
- C) ->
- D) *

Answer: C

197. What will be the output?

C++ MCQs

```
#include <iostream>
using namespace std;

class A {
public:
    virtual void show() {
        cout << "A";
    }
};

class B : public A {
public:
    void show() {
        cout << "B";
    }
};

int main() {
    A *ptr = new B();
    ptr->show();
}
```

- A) A
- B) B
- C) AB
- D) Error

Answer: B

198. Which operator is used for scope resolution?

- A) .
- B) ->
- C) ::
- D) *

Answer: C

199. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 3;

    for(int i = 0; i < 3; i++) {
        x++;
    }

    cout << x;
}
```

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- A) 3
- B) 4
- C) 5
- D) 6

Answer: D

200. Which feature supports code reusability in C++?

- A) Templates
- B) Inheritance
- C) Polymorphism
- D) Encapsulation

Answer: B

201. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;
    cout << x++ << " " << ++x;
}
```

- A) 10 11
- B) 10 12
- C) 11 12
- D) Undefined Behavior

Answer: B

202. Which of the following is not a valid access specifier in C++?

- A) public
- B) private
- C) protected
- D) internal

Answer: D

203. What will be the output?

C++ MCQs

```
#include <iostream>
using namespace std;

class Test {
public:
    Test() {
        cout << "A";
    }
    Test(int x) {
        cout << "B";
    }
};

int main() {
    Test t1;
    Test t2(10);
}
```

- A) AB
- B) BA
- C) AA
- D) BB

Answer: A

204. Which operator cannot be overloaded?

- A) +
- B) []
- C) ::
- D) ==

Answer: C

205. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 20, b = 5;

    cout << a / b * 2;
}
```

- A) 4
- B) 8
- C) 10
- D) 16

C++ MCQs

Answer: B

206. Which keyword is used to refer current object?

- A) object
- B) self
- C) current
- D) this

Answer: D

207. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int arr[3] = {5,10,15};

    cout << *(arr + 1);
}
```

- A) 5
- B) 10
- C) 15
- D) Error

Answer: B

208. Which function is invoked automatically during object copying?

- A) Destructor
- B) Assignment Operator
- C) Copy Constructor
- D) Friend Function

Answer: C

209. What will be the output?

```
#include <iostream>
using namespace std;

class A {
```

C++ MCQs

```
public:
    A() {
        cout << "1";
    }
    ~A() {
        cout << "2";
    }
};

int main() {
    A obj;
}
```

- A) 12
- B) 21
- C) 11
- D) 22

Answer: A

210. Which loop executes at least once?

- A) for
- B) while
- C) do-while
- D) nested loop

Answer: C

211. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 3;

    for(int i = 0; i < 2; i++) {
        x += 2;
    }

    cout << x;
}
```

- A) 5
- B) 6
- C) 7
- D) 8

Answer: C

C++ MCQs

212. Which keyword is used for inheritance with protected access?

- A) protected
- B) private
- C) public
- D) virtual

Answer: A

213. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    static int count;
};

int Test::count = 25;

int main() {
    cout << Test::count;
}
```

- A) 0
- B) 25
- C) Garbage Value
- D) Error

Answer: B

214. Which principle combines variables and functions into one unit?

- A) Abstraction
- B) Encapsulation
- C) Inheritance
- D) Polymorphism

Answer: B

215. What will be the output?

```
#include <iostream>
using namespace std;
```

C++ MCQs

```
int main() {  
    char ch = 'C';  
  
    cout << int(ch);  
}
```

- A) 65
- B) 66
- C) 67
- D) 68

Answer: C

216. Which keyword is used to create dynamic memory?

- A) alloc
- B) malloc
- C) new
- D) create

Answer: C

217. What will be the output?

```
#include <iostream>  
using namespace std;  
  
class Test {  
public:  
    void show() {  
        cout << "Function";  
    }  
};  
  
int main() {  
    Test t;  
    t.show();  
}
```

- A) Function
- B) Error
- C) No Output
- D) Garbage Value

Answer: A

C++ MCQs

218. Which concept allows same function name with different implementations?

- A) Overloading
- B) Encapsulation
- C) Abstraction
- D) Namespace

Answer: A

219. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 4;

    cout << (x > 2 || x < 1);
}
```

- A) 0
- B) 1
- C) TrueFalse
- D) Error

Answer: B

220. Which type of inheritance has one base and many derived classes?

- A) Single
- B) Multiple
- C) Hierarchical
- D) Hybrid

Answer: C

221. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 2;

    switch(x) {
        case 1:
            cout << "One";
    }
```

C++ MCQs

```
        break;

    case 2:
        cout << "Two";
        break;

    default:
        cout << "Default";
    }
}
```

- A) One
- B) Two
- C) Default
- D) Error

Answer: B

222. Which operator is used for stream extraction?

- A) <<
- B) >>
- C) ==
- D) &&

Answer: B

223. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "Constructor";
    }
};

int main() {
    A obj;
}
```

- A) Constructor
- B) Destructor
- C) Error
- D) No Output

Answer: A

C++ MCQs

224. Which STL container allows fast insertion/removal at both ends?

- A) vector
- B) deque
- C) map
- D) set

Answer: B

225. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 11;

    if(x % 2 == 0)
        cout << "Even";
    else
        cout << "Odd";
}
```

- A) Even
- B) Odd
- C) Error
- D) No Output

Answer: B

226. Which operator returns memory address?

- A) *
- B) &
- C) %
- D) #

Answer: B

227. What will be the output?

```
#include <iostream>
using namespace std;

class Base {
```

C++ MCQs

```
public:
    virtual void display() {
        cout << "Base";
    }
};

class Derived : public Base {
public:
    void display() {
        cout << "Derived";
    }
};

int main() {
    Base *ptr = new Derived;
    ptr->display();
}
```

- A) Base
- B) Derived
- C) Error
- D) No Output

Answer: B

228. Which keyword is used for immutable variables?

- A) static
- B) final
- C) const
- D) immutable

Answer: C

229. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;

    while(x > 2) {
        cout << x;
        x--;
    }
}
```

- A) 543
- B) 54
- C) 321

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- D) Infinite Loop

Answer: A

230. Which mechanism is used for error handling in C++?

- A) goto
- B) exceptions
- C) loops
- D) templates

Answer: B

231. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "A";
    }
};

class B : public A {
public:
    B() {
        cout << "B";
    }
};

class C : public B {
public:
    C() {
        cout << "C";
    }
};

int main() {
    C obj;
}
```

- A) ABC
- B) CBA
- C) BAC
- D) ACB

Answer: A

C++ MCQs

232. Which function opens a file in C++ streams?

- A) fopen()
- B) open()
- C) start()
- D) begin()

Answer: B

233. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 6;

    cout << x * 3;
}
```

- A) 9
- B) 12
- C) 18
- D) 24

Answer: C

234. Which storage class creates local variable only once?

- A) auto
- B) register
- C) static
- D) extern

Answer: C

235. What will be the output?

```
#include <iostream>
using namespace std;

template <typename T>
T multiply(T a, T b) {
    return a * b;
}

int main() {
```


C++ MCQs

```
    cout << multiply(3,4);  
}
```

- A) 7
- B) 12
- C) 34
- D) Error

Answer: B

236. Which keyword avoids naming conflicts?

- A) class
- B) namespace
- C) template
- D) friend

Answer: B

237. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int x = 1;  
  
    do {  
        cout << x;  
        x++;  
    } while(x <= 3);  
}
```

- A) 123
- B) 0123
- C) 321
- D) Infinite Loop

Answer: A

238. Which operator compares two values?

- A) =
- B) ==
- C) +=
- D) %=

C++ MCQs

Answer: B

239. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    static void display() {
        cout << "Static";
    }
};

int main() {
    Test::display();
}
```

- A) Static
- B) Error
- C) No Output
- D) Garbage Value

Answer: A

240. Which constructor initializes object using another object?

- A) Default Constructor
- B) Parameterized Constructor
- C) Copy Constructor
- D) Virtual Constructor

Answer: C

241. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 17;
    int b = 5;

    cout << a % b;
}
```

- A) 1
- B) 2

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- C) 3
- D) 4

Answer: B

242. Which keyword restricts function overriding?

- A) const
- B) virtual
- C) final
- D) static

Answer: C

243. What will be the output?

```
#include <iostream>
using namespace std;

class Demo {
public:
    int x = 100;
};

int main() {
    Demo d;
    cout << d.x;
}
```

- A) 0
- B) 10
- C) 100
- D) Garbage Value

Answer: C

244. Which header file is required for vectors?

- A) queue
- B) stack
- C) vector
- D) list

Answer: C

C++ MCQs

245. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;

    cout << sizeof(char);
}
```

- A) 1
- B) 2
- C) 4
- D) Depends on compiler

Answer: A

246. Which operator represents logical NOT?

- A) &&
- B) ||
- C) !
- D) !=

Answer: C

247. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    virtual void show() {
        cout << "A";
    }
};

class B : public A {
public:
    void show() {
        cout << "B";
    }
};

int main() {
    B b;
    A &ref = b;
```

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```
        ref.show();  
    }
```

- A) A
- B) B
- C) AB
- D) Error

Answer: B

248. Which operator can be overloaded for custom indexing?

- A) ()
- B) []
- C) {}
- D) <>

Answer: B

249. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int sum = 0;  
  
    for(int i = 1; i <= 3; i++) {  
        sum += i;  
    }  
  
    cout << sum;  
}
```

- A) 3
- B) 5
- C) 6
- D) 7

Answer: C

250. Which concept supports reusability of existing classes?

- A) Abstraction
- B) Inheritance
- C) Encapsulation

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- D) Templates

Answer: B

251. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    Test() {
        cout << "Start ";
    }

    ~Test() {
        cout << "End";
    }
};

int main() {
    Test obj;
}
```

- A) Start
- B) End
- C) Start End
- D) End Start

Answer: C

252. Which STL container stores unique sorted elements?

- A) vector
- B) queue
- C) set
- D) stack

Answer: C

253. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 9;
```

C++ MCQs

```
    cout << (x > 5 ? x : 0);  
}
```

- A) 0
- B) 5
- C) 9
- D) Error

Answer: C

254. Which keyword grants non-member function access to private members?

- A) protected
- B) private
- C) friend
- D) public

Answer: C

255. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int arr[3] = {1,2,3};  
  
    cout << arr[0] + arr[2];  
}
```

- A) 3
- B) 4
- C) 5
- D) 6

Answer: B

256. Which cast safely converts within inheritance hierarchy?

- A) reinterpret_cast
- B) dynamic_cast
- C) const_cast
- D) static_cast

Answer: B

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257. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;

    x -= 3;

    cout << x;
}
```

- A) 5
- B) 6
- C) 7
- D) 8

Answer: C

258. Which polymorphism is achieved using virtual functions?

- A) Compile-time
- B) Runtime
- C) Static
- D) Early Binding

Answer: B

259. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 3;

    switch(x) {
        case 1:
            cout << "A";
            break;

        case 3:
            cout << "B";
            break;
    }
}
```

- A) A

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- B) B
- C) AB
- D) Error

Answer: B

260. Which keyword suggests function expansion inline?

- A) static
- B) inline
- C) virtual
- D) friend

Answer: B

261. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;
    cout << x++ << " " << ++x;
}
```

- A) 10 11
- B) 10 12
- C) 11 12
- D) Undefined Behavior

Answer: B

262. Which of the following is not a valid access specifier in C++?

- A) public
- B) private
- C) protected
- D) internal

Answer: D

263. What will be the output?

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```
#include <iostream>
using namespace std;

class Test {
public:
    Test() {
        cout << "A";
    }
    Test(int x) {
        cout << "B";
    }
};

int main() {
    Test t1;
    Test t2(10);
}
```

- A) AB
- B) BA
- C) AA
- D) BB

Answer: A

264. Which operator cannot be overloaded?

- A) +
- B) []
- C) ::
- D) ==

Answer: C

265. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 20, b = 5;

    cout << a / b * 2;
}
```

- A) 4
- B) 8
- C) 10
- D) 16

C++ MCQs

Answer: B

266. Which keyword is used to refer current object?

- A) object
- B) self
- C) current
- D) this

Answer: D

267. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int arr[3] = {5,10,15};

    cout << *(arr + 1);
}
```

- A) 5
- B) 10
- C) 15
- D) Error

Answer: B

268. Which function is invoked automatically during object copying?

- A) Destructor
- B) Assignment Operator
- C) Copy Constructor
- D) Friend Function

Answer: C

269. What will be the output?

```
#include <iostream>
using namespace std;

class A {
```

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```
public:
    A() {
        cout << "1";
    }
    ~A() {
        cout << "2";
    }
};

int main() {
    A obj;
}
```

- A) 12
- B) 21
- C) 11
- D) 22

Answer: A

270. Which loop executes at least once?

- A) for
- B) while
- C) do-while
- D) nested loop

Answer: C

271. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 3;

    for(int i = 0; i < 2; i++) {
        x += 2;
    }

    cout << x;
}
```

- A) 5
- B) 6
- C) 7
- D) 8

Answer: C

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272. Which keyword is used for inheritance with protected access?

- A) protected
- B) private
- C) public
- D) virtual

Answer: A

273. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    static int count;
};

int Test::count = 25;

int main() {
    cout << Test::count;
}
```

- A) 0
- B) 25
- C) Garbage Value
- D) Error

Answer: B

274. Which principle combines variables and functions into one unit?

- A) Abstraction
- B) Encapsulation
- C) Inheritance
- D) Polymorphism

Answer: B

275. What will be the output?

```
#include <iostream>
using namespace std;
```

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```
int main() {  
    char ch = 'C';  
  
    cout << int(ch);  
}
```

- A) 65
- B) 66
- C) 67
- D) 68

Answer: C

276. Which keyword is used to create dynamic memory?

- A) alloc
- B) malloc
- C) new
- D) create

Answer: C

277. What will be the output?

```
#include <iostream>  
using namespace std;  
  
class Test {  
public:  
    void show() {  
        cout << "Function";  
    }  
};  
  
int main() {  
    Test t;  
    t.show();  
}
```

- A) Function
- B) Error
- C) No Output
- D) Garbage Value

Answer: A

C++ MCQs

278. Which concept allows same function name with different implementations?

- A) Overloading
- B) Encapsulation
- C) Abstraction
- D) Namespace

Answer: A

279. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 4;

    cout << (x > 2 || x < 1);
}
```

- A) 0
- B) 1
- C) TrueFalse
- D) Error

Answer: B

280. Which type of inheritance has one base and many derived classes?

- A) Single
- B) Multiple
- C) Hierarchical
- D) Hybrid

Answer: C

281. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 2;

    switch(x) {
        case 1:
            cout << "One";
    }
```

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```
        break;

    case 2:
        cout << "Two";
        break;

    default:
        cout << "Default";
    }
}
```

- A) One
- B) Two
- C) Default
- D) Error

Answer: B

282. Which operator is used for stream extraction?

- A) <<
- B) >>
- C) ==
- D) &&

Answer: B

283. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "Constructor";
    }
};

int main() {
    A obj;
}
```

- A) Constructor
- B) Destructor
- C) Error
- D) No Output

Answer: A

C++ MCQs

284. Which STL container allows fast insertion/removal at both ends?

- A) vector
- B) deque
- C) map
- D) set

Answer: B

285. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 11;

    if(x % 2 == 0)
        cout << "Even";
    else
        cout << "Odd";
}
```

- A) Even
- B) Odd
- C) Error
- D) No Output

Answer: B

286. Which operator returns memory address?

- A) *
- B) &
- C) %
- D) #

Answer: B

287. What will be the output?

```
#include <iostream>
using namespace std;

class Base {
```

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```
public:
    virtual void display() {
        cout << "Base";
    }
};

class Derived : public Base {
public:
    void display() {
        cout << "Derived";
    }
};

int main() {
    Base *ptr = new Derived;
    ptr->display();
}
```

- A) Base
- B) Derived
- C) Error
- D) No Output

Answer: B

288. Which keyword is used for immutable variables?

- A) static
- B) final
- C) const
- D) immutable

Answer: C

289. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;

    while(x > 2) {
        cout << x;
        x--;
    }
}
```

- A) 543
- B) 54
- C) 321

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- D) Infinite Loop

Answer: A

290. Which mechanism is used for error handling in C++?

- A) goto
- B) exceptions
- C) loops
- D) templates

Answer: B

291. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    A() {
        cout << "A";
    }
};

class B : public A {
public:
    B() {
        cout << "B";
    }
};

class C : public B {
public:
    C() {
        cout << "C";
    }
};

int main() {
    C obj;
}
```

- A) ABC
- B) CBA
- C) BAC
- D) ACB

Answer: A

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292. Which function opens a file in C++ streams?

- A) fopen()
- B) open()
- C) start()
- D) begin()

Answer: B

293. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 6;

    cout << x * 3;
}
```

- A) 9
- B) 12
- C) 18
- D) 24

Answer: C

294. Which storage class creates local variable only once?

- A) auto
- B) register
- C) static
- D) extern

Answer: C

295. What will be the output?

```
#include <iostream>
using namespace std;

template <typename T>
T multiply(T a, T b) {
    return a * b;
}

int main() {
```

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```
    cout << multiply(3,4);  
}
```

- A) 7
- B) 12
- C) 34
- D) Error

Answer: B

296. Which keyword avoids naming conflicts?

- A) class
- B) namespace
- C) template
- D) friend

Answer: B

297. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int x = 1;  
  
    do {  
        cout << x;  
        x++;  
    } while(x <= 3);  
}
```

- A) 123
- B) 0123
- C) 321
- D) Infinite Loop

Answer: A

298. Which operator compares two values?

- A) =
- B) ==
- C) +=
- D) %=

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Answer: B

299. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    static void display() {
        cout << "Static";
    }
};

int main() {
    Test::display();
}
```

- A) Static
- B) Error
- C) No Output
- D) Garbage Value

Answer: A

300. Which constructor initializes object using another object?

- A) Default Constructor
- B) Parameterized Constructor
- C) Copy Constructor
- D) Virtual Constructor

Answer: C

301. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int a = 17;
    int b = 5;

    cout << a % b;
}
```

- A) 1
- B) 2

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- C) 3
- D) 4

Answer: B

302. Which keyword restricts function overriding?

- A) const
- B) virtual
- C) final
- D) static

Answer: C

303. What will be the output?

```
#include <iostream>
using namespace std;

class Demo {
public:
    int x = 100;
};

int main() {
    Demo d;
    cout << d.x;
}
```

- A) 0
- B) 10
- C) 100
- D) Garbage Value

Answer: C

304. Which header file is required for vectors?

- A) queue
- B) stack
- C) vector
- D) list

Answer: C

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305. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 5;

    cout << sizeof(char);
}
```

- A) 1
- B) 2
- C) 4
- D) Depends on compiler

Answer: A

306. Which operator represents logical NOT?

- A) &&
- B) ||
- C) !
- D) !=

Answer: C

307. What will be the output?

```
#include <iostream>
using namespace std;

class A {
public:
    virtual void show() {
        cout << "A";
    }
};

class B : public A {
public:
    void show() {
        cout << "B";
    }
};

int main() {
    B b;
    A &ref = b;
```


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```
        ref.show();  
    }
```

- A) A
- B) B
- C) AB
- D) Error

Answer: B

308. Which operator can be overloaded for custom indexing?

- A) ()
- B) []
- C) {}
- D) <>

Answer: B

309. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int sum = 0;  
  
    for(int i = 1; i <= 3; i++) {  
        sum += i;  
    }  
  
    cout << sum;  
}
```

- A) 3
- B) 5
- C) 6
- D) 7

Answer: C

310. Which concept supports reusability of existing classes?

- A) Abstraction
- B) Inheritance
- C) Encapsulation

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- D) Templates

Answer: B

311. What will be the output?

```
#include <iostream>
using namespace std;

class Test {
public:
    Test() {
        cout << "Start ";
    }

    ~Test() {
        cout << "End";
    }
};

int main() {
    Test obj;
}
```

- A) Start
- B) End
- C) Start End
- D) End Start

Answer: C

312. Which STL container stores unique sorted elements?

- A) vector
- B) queue
- C) set
- D) stack

Answer: C

313. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 9;
```

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```
    cout << (x > 5 ? x : 0);  
}
```

- A) 0
- B) 5
- C) 9
- D) Error

Answer: C

314. Which keyword grants non-member function access to private members?

- A) protected
- B) private
- C) friend
- D) public

Answer: C

315. What will be the output?

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int arr[3] = {1,2,3};  
  
    cout << arr[0] + arr[2];  
}
```

- A) 3
- B) 4
- C) 5
- D) 6

Answer: B

316. Which cast safely converts within inheritance hierarchy?

- A) reinterpret_cast
- B) dynamic_cast
- C) const_cast
- D) static_cast

Answer: B

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317. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;

    x -= 3;

    cout << x;
}
```

- A) 5
- B) 6
- C) 7
- D) 8

Answer: C

318. Which polymorphism is achieved using virtual functions?

- A) Compile-time
- B) Runtime
- C) Static
- D) Early Binding

Answer: B

319. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    int x = 3;

    switch(x) {
        case 1:
            cout << "A";
            break;

        case 3:
            cout << "B";
            break;
    }
}
```

- A) A

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- B) B
- C) AB
- D) Error

Answer: B

320. Which keyword suggests function expansion inline?

- A) static
- B) inline
- C) virtual
- D) friend

Answer: B
